Student Food Spending Classification Using K-NN

A university wants to understand and classify students based on their **daily food spending behavior**.

They already collected student data (budget, meals, snacks, drinks, distance), but they want a way to **automatically classify new students** into:

- Saver
- Balanced
- Spender

DATA:

ID	DailyFoodBudget	MealsPerDay	SnackSpending	DrinksSpending	DistanceToCampus	Class
1	120	2	30	20	1.2	Saver
2	150	2	40	25	0.8	Saver
3	180	2	50	30	1.5	Saver
4	200	3	60	35	2.0	Saver
5	220	3	70	40	1.0	Saver
6	250	3	80	45	1.8	Balanced
7	280	3	90	50	2.2	Balanced
8	300	3	100	60	2.5	Balanced
9	320	4	110	70	1.6	Balanced
10	350	4	120	75	2.8	Balanced
11	380	4	130	80	3.0	Spender
12	400	4	140	85	3.2	Spender
13	420	5	150	90	2.0	Spender
14	450	5	160	100	2.5	Spender
15	480	5	170	110	3.5	Spender

16	500	5	180	120	4.0	Spender
17	130	2	35	20	1.0	Saver
18	160	2	45	25	1.3	Saver
19	210	3	65	35	1.7	Saver
20	260	3	85	50	2.1	Balanced
21	290	3	95	55	2.4	Balanced
22	310	4	105	65	2.7	Balanced
23	340	4	115	70	2.9	Balanced
24	390	4	125	80	3.1	Spender
25	410	5	135	85	3.3	Spender
26	430	5	145	95	3.6	Spender
27	460	5	155	105	3.8	Spender
28	490	5	165	115	4.1	Spender
29	520	6	180	130	4.5	Spender
30	550	6	190	140	5.0	Spender

Steps

1. Choose K Value
K = 3

2. New Data (Test Point)

Example student:

Budget = 275
Meals = 3
Snacks = 85
Drinks = 50

Distance = 2.0

3. Compute Distance (Euclidean Distance)

Formula:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2 + (a_1 - a_2)^2 + (b_1 - b_2)^2}$$

Apply this across all features:

$d = \sqrt{[(Budget_1 - Budget_2)^2 + (Meals_1 - Meals_2)^2 + (Snacks_1 - Snacks_2)^2 + (Drinks_1 - Drinks_2)^2 + (Distance_1 - Distance_2)^2]}$

Step A: Subtract the values (Finding the differences)

- Budget: 120 - 275 = -155
- Meals: 2 - 3 = -1
- Snacks: 30 - 85 = -55
- Drinks: 20 - 50 = -30
- Distance: 1.2 - 2.0 = -0.8

Step B: Square each difference

- (-155)^2 = 24,025
- (-1)^2 = 1
- (-55)^2 = 3,025
- (-30)^2 = 900
- (-0.8)^2 = 0.64

Step C: Sum the squared values

Sum = 24,025 + 1 + 3,025 + 900 + 0.64
Sum = 27,951.64

Step D: Take the Square Root

$$d = \sqrt{27,951.64}$$

$$d \approx 167.1874397$$

COMPUTED DISTANCE:

ID	Class	Distance
1	Saver	167.18
2	Saver	135.19
3	Saver	98.62
4	Saver	75.50
5	Saver	53.21
6	Balanced	25.98
7	Balanced	**7.07**
8	Balanced	30.82
9	Balanced	50.99
10	Balanced	75.83
11	Spender	101.61
12	Spender	121.84
13	Spender	144.25
14	Spender	176.99
15	Spender	209.82
16	Spender	237.17
17	Saver	153.53
18	Saver	120.56
19	Saver	65.38
20	Balanced	**15.00**
21	Balanced	**18.71**
22	Balanced	40.35
23	Balanced	66.37
24	Spender	111.92
25	Spender	131.23
26	Spender	157.42

27	Spender	183.85
28	Spender	210.39
29	Spender	254.51
30	Spender	307.84

4. Find Nearest Neighbors

ID	Distance	Class
7	**7.07**	Balanced
20	**15.00**	Balanced
21	**18.71**	Balanced

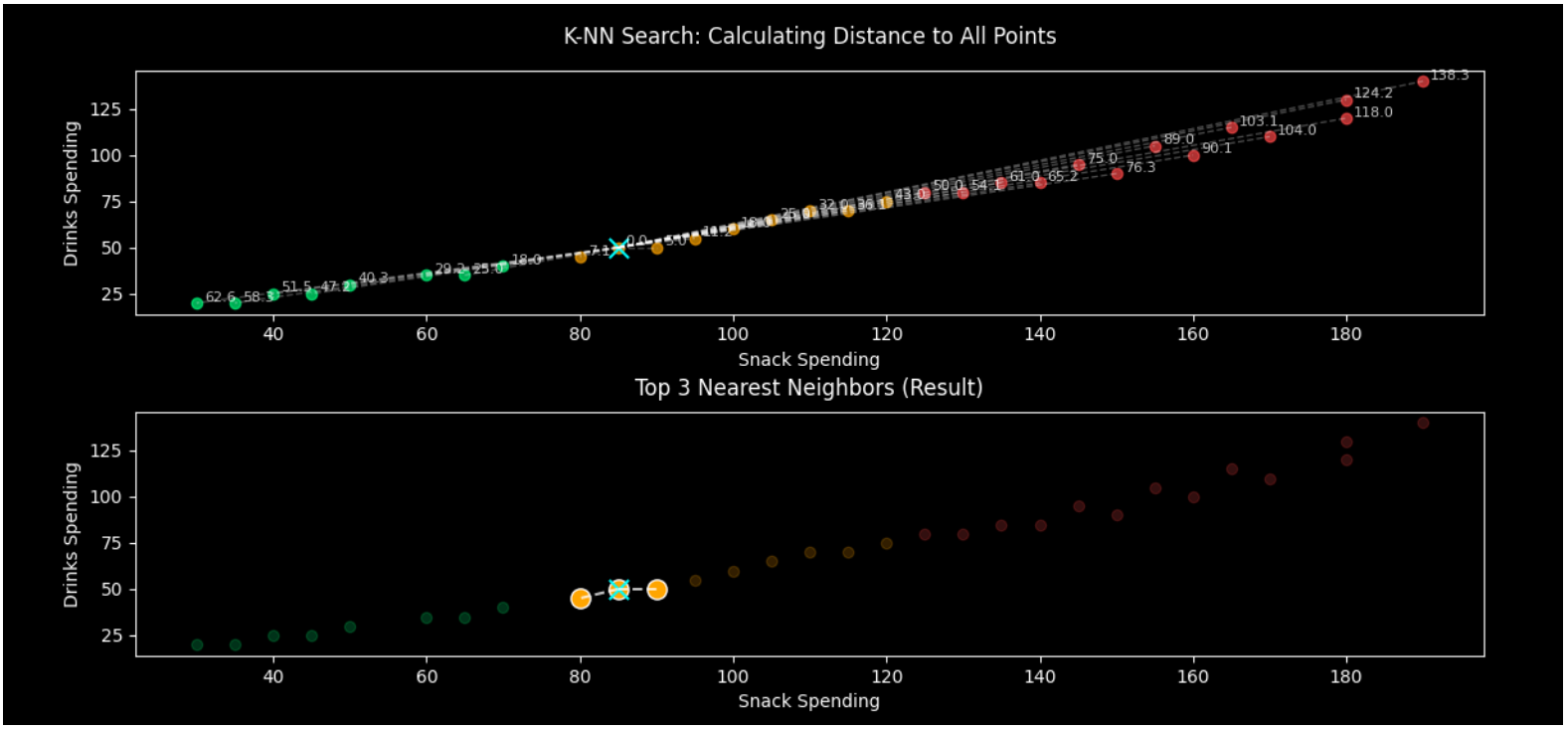
5. Classification (Majority Voting)
Balanced = 3
Saver = 0
Spender = 0

Predicted Class = BALANCED

6. Visualization

Plot:

- Budget vs TotalSpending OR Snacks vs Drinks
- Highlight:
    - Existing data points
    - New data point
    - Neighbor connections


K-NN Search: Calculating Distance to All Points
Drinks Spending
Snack Spending
Top 3 Nearest Neighbors (Result)
Drinks Spending
Snack Spending

KMEANS Dataset

Cluster 1: saver

Cluster 2: balanced

Cluster 3: spender

ID	Budget	Meals	Snacks	Drinks
1	120	2	30	20
2	150	2	40	25
3	180	2	50	30
4	200	3	60	35
5	220	3	70	40
6	250	3	80	45
7	280	3	90	50
8	300	3	100	60
9	320	4	110	70
10	350	4	120	75
11	380	4	130	80
12	400	4	140	85
13	420	5	150	90
14	450	5	160	100
15	480	5	170	110
16	500	5	180	120
17	130	2	35	20
18	160	2	45	25
19	210	3	65	35
20	260	3	85	50
21	290	3	95	55
22	310	4	105	65
23	340	4	115	70
24	390	4	125	80

25	410	5	135	85
26	430	5	145	95
27	460	5	155	105
28	490	5	165	115
29	520	6	180	130
30	550	6	190	140

Solve for total spending

TotalSpending=Snacks+Drinks

ID	Snacks	Drinks	Calculation (S+D)	Total Spending
1	30	20	30 + 20	**50**
2	40	25	40 + 25	**65**
3	50	30	50 + 30	**80**
4	60	35	60 + 35	**95**
5	70	40	70 + 40	**110**
6	80	45	80 + 45	**125**
7	90	50	90 + 50	**140**
8	100	60	100 + 60	**160**
9	110	70	110 + 70	**180**
10	120	75	120 + 75	**195**
11	130	80	130 + 80	**210**
12	140	85	140 + 85	**225**
13	150	90	150 + 90	**240**
14	160	100	160 + 100	**260**
15	170	110	170 + 110	**280**
16	180	120	180 + 120	**300**

Step 1: Choose the number of clusters (K)

Based on the Elbow Method previously discussed, we choose **K=3**.

Step 2: Choose K Centroids at Random

Let's pick three starting points (centroids) from our dataset to represent our initial guesses for C1, C2, and C3:

- **Centroid 1 (m1):** ID 1 (Budget: 120, Spending: 50)
- **Centroid 2 (m2):** ID 10 (Budget: 350, Spending: 195)
- **Centroid 3 (m3):** ID 30 (Budget: 550, Spending: 330)

Step 3: Assign each data point to the closest Centroid

We use the **Euclidean Distance** formula to find which centroid is "nearest" to each ID

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example solving for ID 3 (Budget: 180, Spending: 80):

1. **Distance to m_1:** $\sqrt{(180-120)^2 + (80-50)^2} = \sqrt{60^2 + 30^2} = \mathbf{67.08}$
2. **Distance to m_2:** $\sqrt{(180-350)^2 + (80-195)^2} = \sqrt{(-170)^2 + (-115)^2} = \mathbf{205.24}$
3. **Distance to m_3:** $\sqrt{(180-550)^2 + (80-330)^2} = \sqrt{(-370)^2 + (-250)^2} = \mathbf{446.54}$

Result: ID 3 is closest to m_1, so it is assigned to **Cluster 1**.

Step 4 & 5: Determine Variance and Reposition Centroids

Solving for the new Centroid 1 (C1):

- **New Budget Mean:** $\frac{\sum \text{Budgets in C1}}{\text{Count of IDs in C1}}$
- **New Spending Mean:** $\frac{\sum \text{Spending in C1}}{\text{Count of IDs in C1}}$

If C1 contains IDs 1, 2, 3, 17, and 18:

- Budget Mean: $(120 + 150 + 180 + 130 + 160)/5 = \mathbf{148}$
- Spending Mean: $(50 + 65 + 80 + 55 + 70)/5 = \mathbf{64}$
- **New m_1 position is (148, 64).**

Step 6: The Iteration Loop

Round 1: The Initial Guess

We pick 3 random points (Initial m) and assign the closest IDs.

- **Initial m1 (C1):** (120, 50) — *Used ID 1 as a starting point.*
- **Initial m2 (C2):** (280, 140) — *Used ID 7 as a starting point.*
- **Initial m3 (C3):** (550, 330) — *Used ID 30 as a starting point.*

Assignment: IDs 1-5 and 17-19 go to C1. IDs 6-10 and 20-23 go to C2. IDs 11-16 and 24-29 go to C3.

Round 2: The First "Learning" Step (Updating m)

Now we calculate the **actual mean** of the IDs assigned to each group. This moves the centroid to the "center of gravity" of those points.

Calculating New m1 (C1):

- **Budget Sum:** 120+150+180+200+220+130+160+210=1,370
- **TotalSpending Sum:** 50+65+80+95+110+55+70+100=625
- **New m1:** (81370,8625)=(171.25,78.12)

Calculating New m2 (C2):

- **Budget Sum:** 250+280+300+320+350+260+290+310+340=2,700
- **TotalSpending Sum:** 125+140+160+180+195+135+150+170+185=1,440
- **New m2:** (92700,91440)=(300,160)

Calculating New m3 (C3):

- **New m3:** (Average of IDs 11-16 and 24-30) = (453,258)

Round 3: Re-Assignment (Step 6)

We check if any ID is now closer to a *different* m.

- **Check ID 11:** (Budget 380, Spending 210).
    - Distance to m2 (300, 160): ≈94
    - Distance to m3 (453, 258): ≈87
- **Decision:** ID 11 stays in C3.

If we check all 30 points and **none of them switch clusters**, we have reached **Convergence**.

Step 7: Finish (Stability)

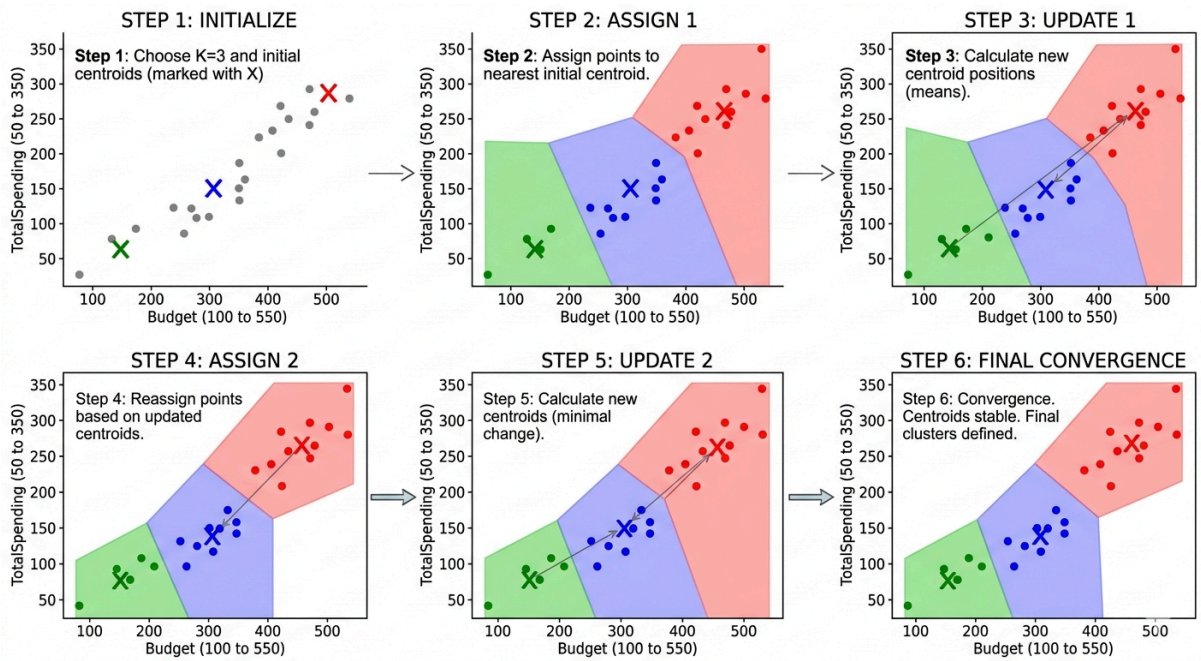
The "solving" is complete when:

1. **Re-assignment results in no change:** Every ID stays in the same cluster color.
2. **m stops moving:** The calculation for the mean produces the exact same coordinates as the previous round.

Cluster	Final m (Budget, Spending)	Meaning
C1	**(171, 78)**	The "Average" Low Spender
C2	**(300, 160)**	The "Average" Mid Spender
C3	**(453, 258)**	The "Average" High Spender

Analysis of Results

- Cluster Profiles: * Cluster 1 (Savers): Characterized by low DailyFoodBudget, fewer MealsPerDay, and shorter DistanceToCampus.
    - Cluster 3 (Spenders): Characterized by high DailyFoodBudget (up to 500), high Snack/Drink spending, and living further from campus.
- Convergence: The process is complete (converged) because, in the final iteration, no student IDs changed their cluster assignment.



Technical Accuracy Note

It is important to note that k-Means is an unsupervised learning algorithm. Even though the "Class" labels (Saver, Balanced, Spender) existed in our raw file, the algorithm remained "blind" to them. It grouped the students based purely on the mathematical proximity of their spending data, proving the labels were accurate.